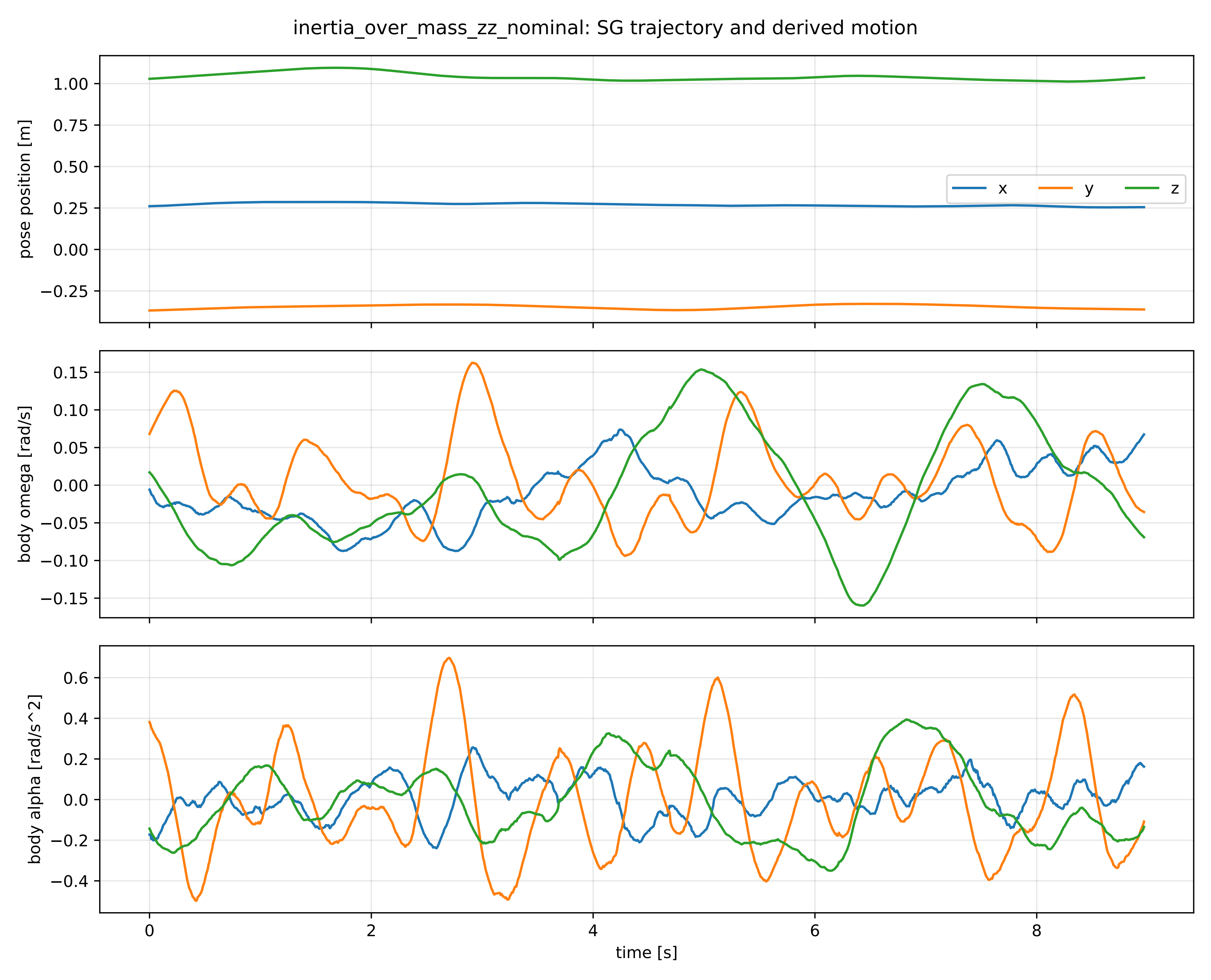
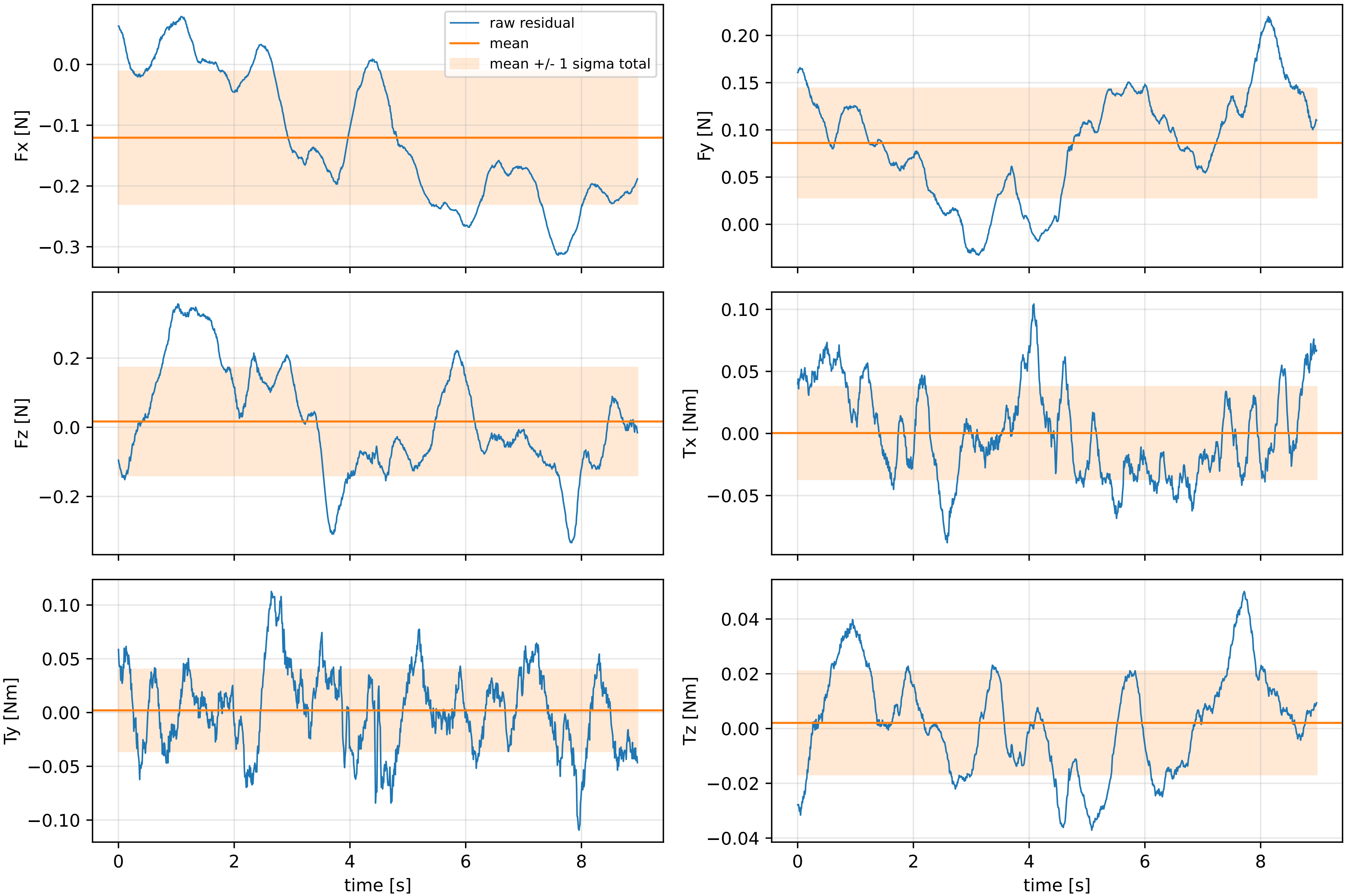
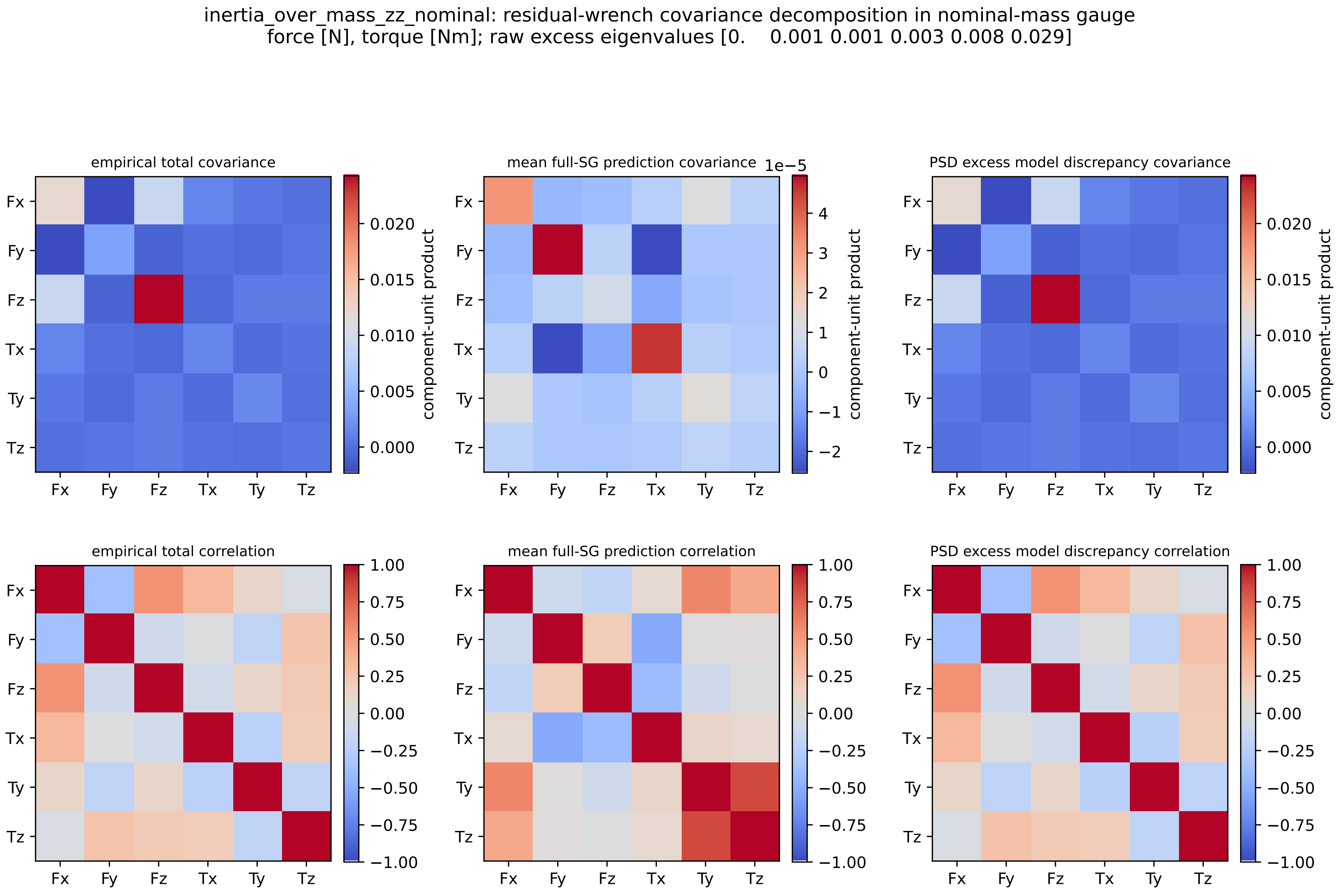
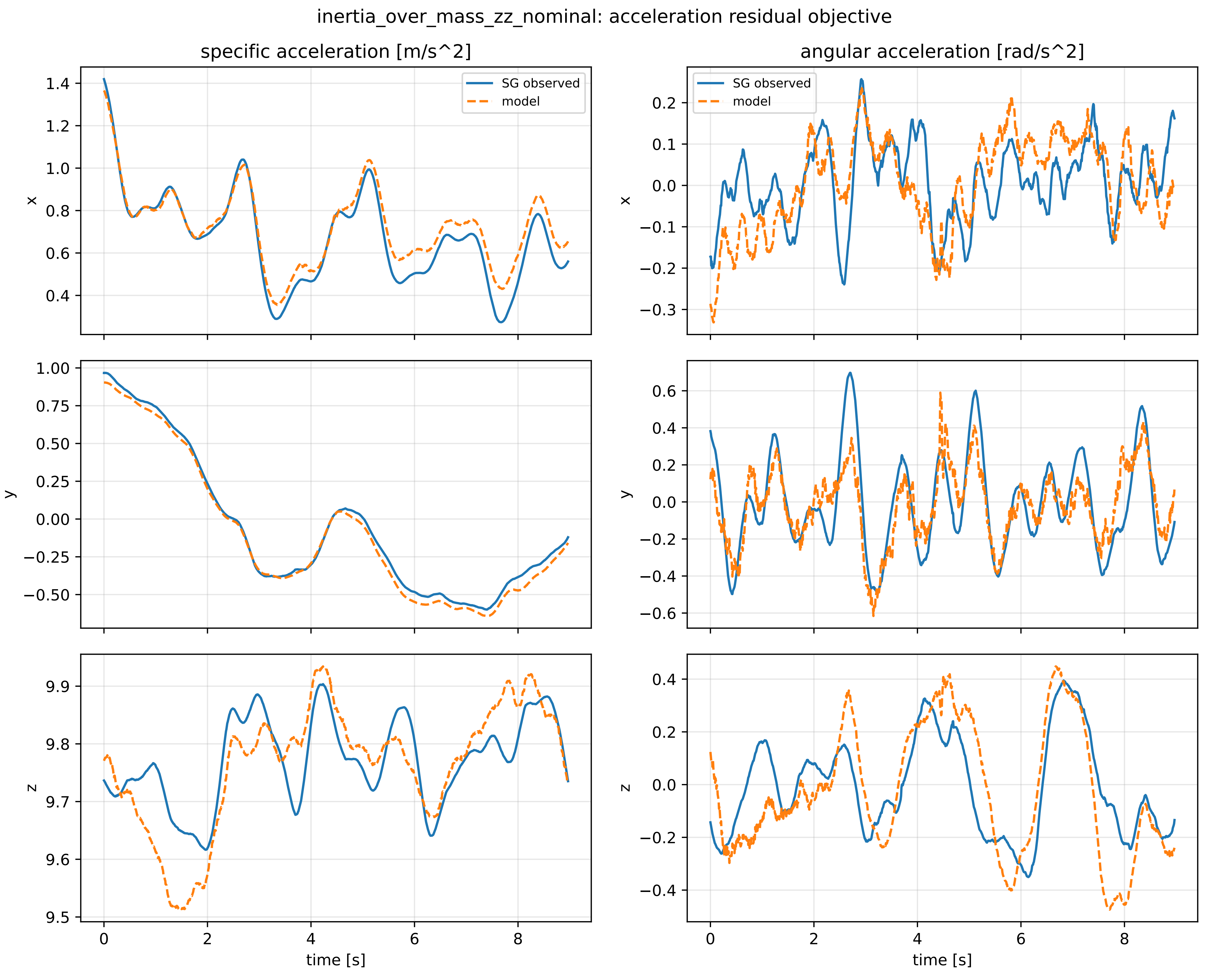

inertia_over_mass_zz_nominal: SG trajectory and derived motion
pose position [m]
body omega [rad/s]
body alpha [rad/s^2]
time [s]
x
y
z
1.00
0.75
0.50
0.25
0.00
−0.25
0.15
0.10
0.05
0.00
−0.05
−0.10
−0.15
0.6
0.4
0.2
0.0
−0.2
−0.4
0
2
4
6
8

inertia_over_mass_zz_nominal: raw Newton--Euler residual wrench (not trajectory-fitted external wrench)
mass gauge fixed to nominal mass = 2.35159759 kg

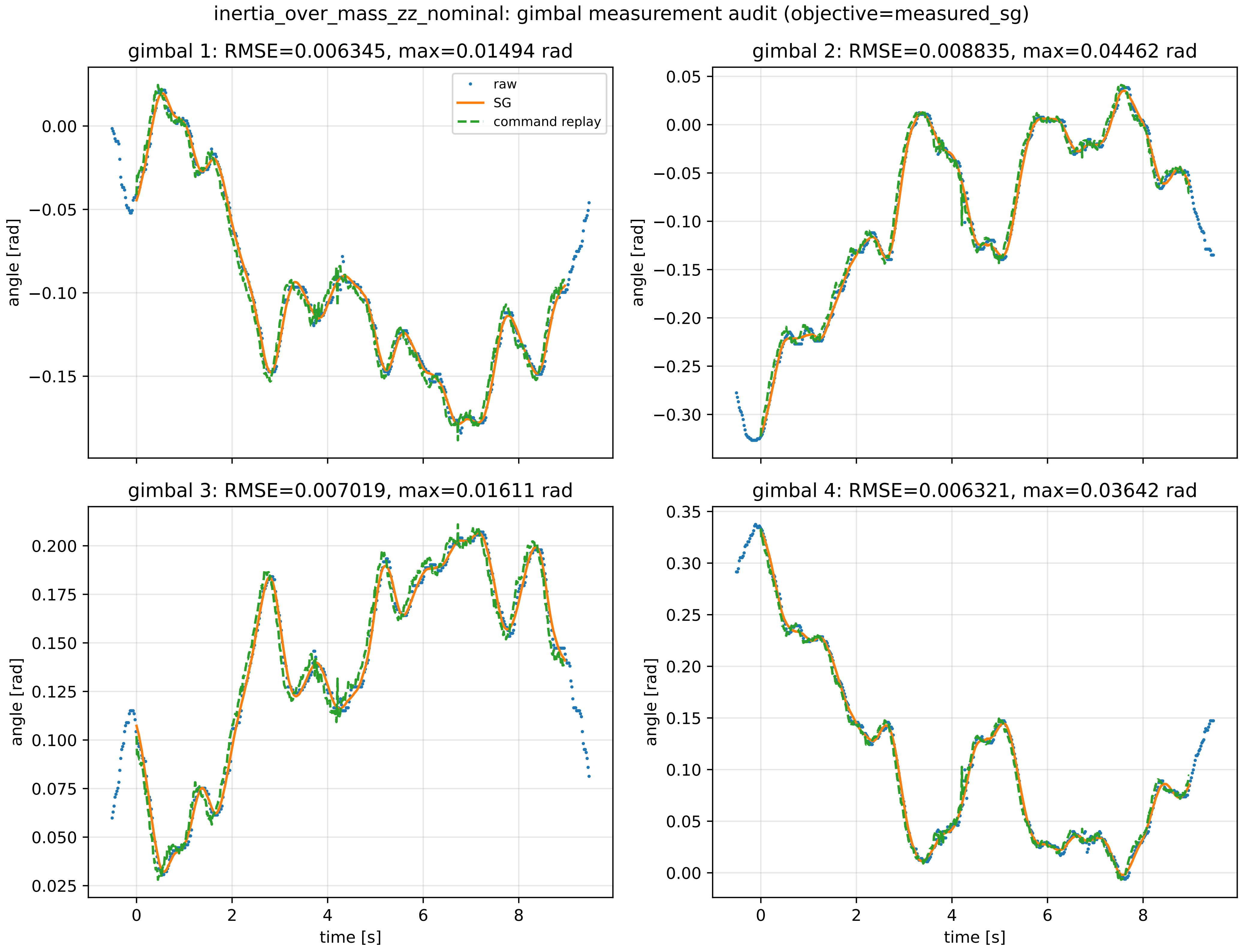


vector RMS about zero: force 0.24847 N, torque 0.056494 Nm; component std about mean: force [0.1091 0.0578 0.1558], torque [0.0372 0.038 0.0188]

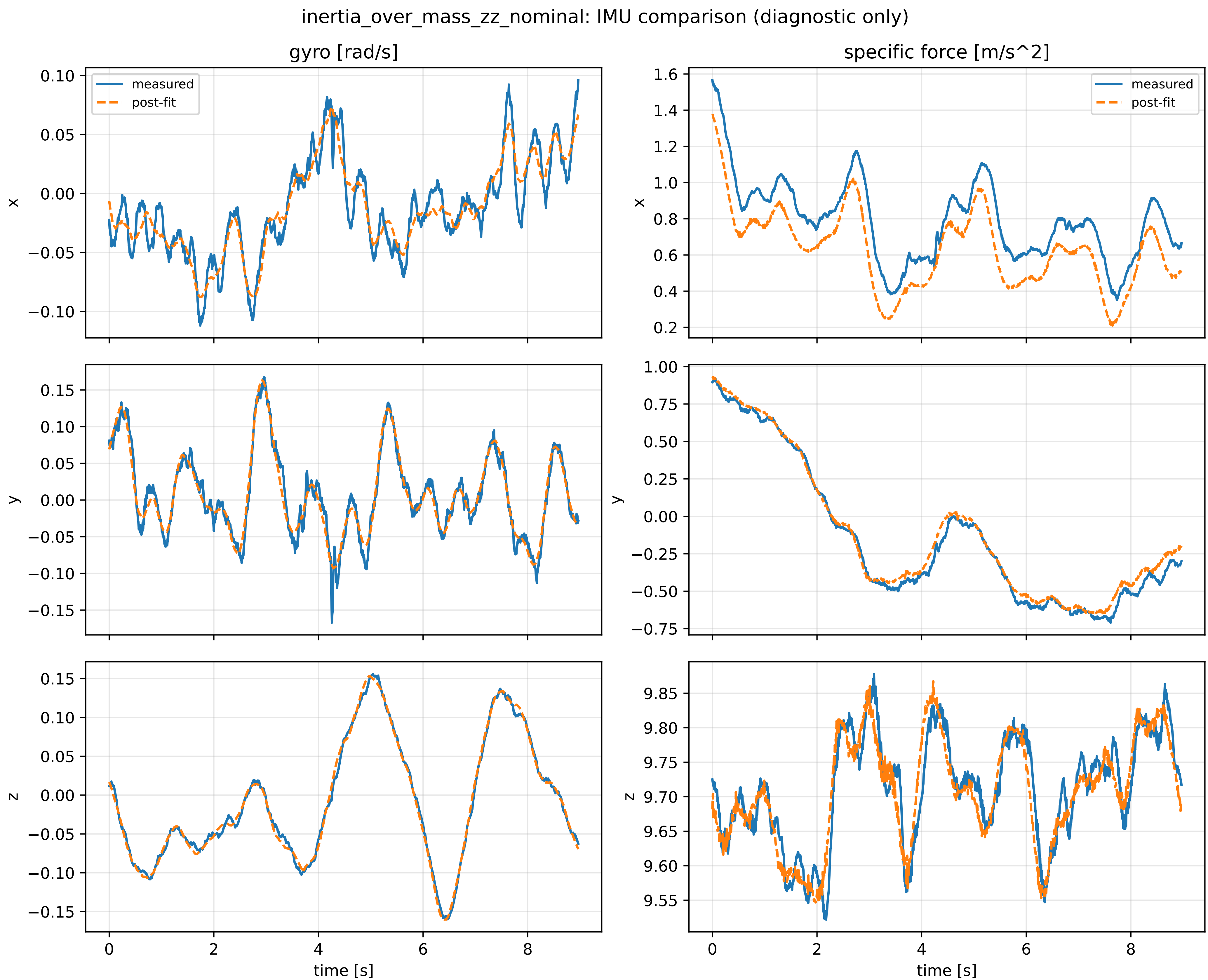

inertia_over_mass_zz_nominal: residual-wrench covariance decomposition in nominal-mass gauge
force [N], torque [Nm]; raw excess eigenvalues [0.   0.001 0.001 0.003 0.008 0.029]
empirical total covariance
mean full-SG prediction covariance
1e−5
PSD excess model discrepancy covariance
empirical total correlation
mean full-SG prediction correlation
PSD excess model discrepancy correlation
component-unit product
Fx
Fy
Fz
Tx
Ty
Tz
0.000
0.005
0.010
0.015
0.020
−2
−1
0
1
2
3
4
−1.00
−0.75
−0.50
−0.25
0.00
0.25
0.50
0.75
1.00

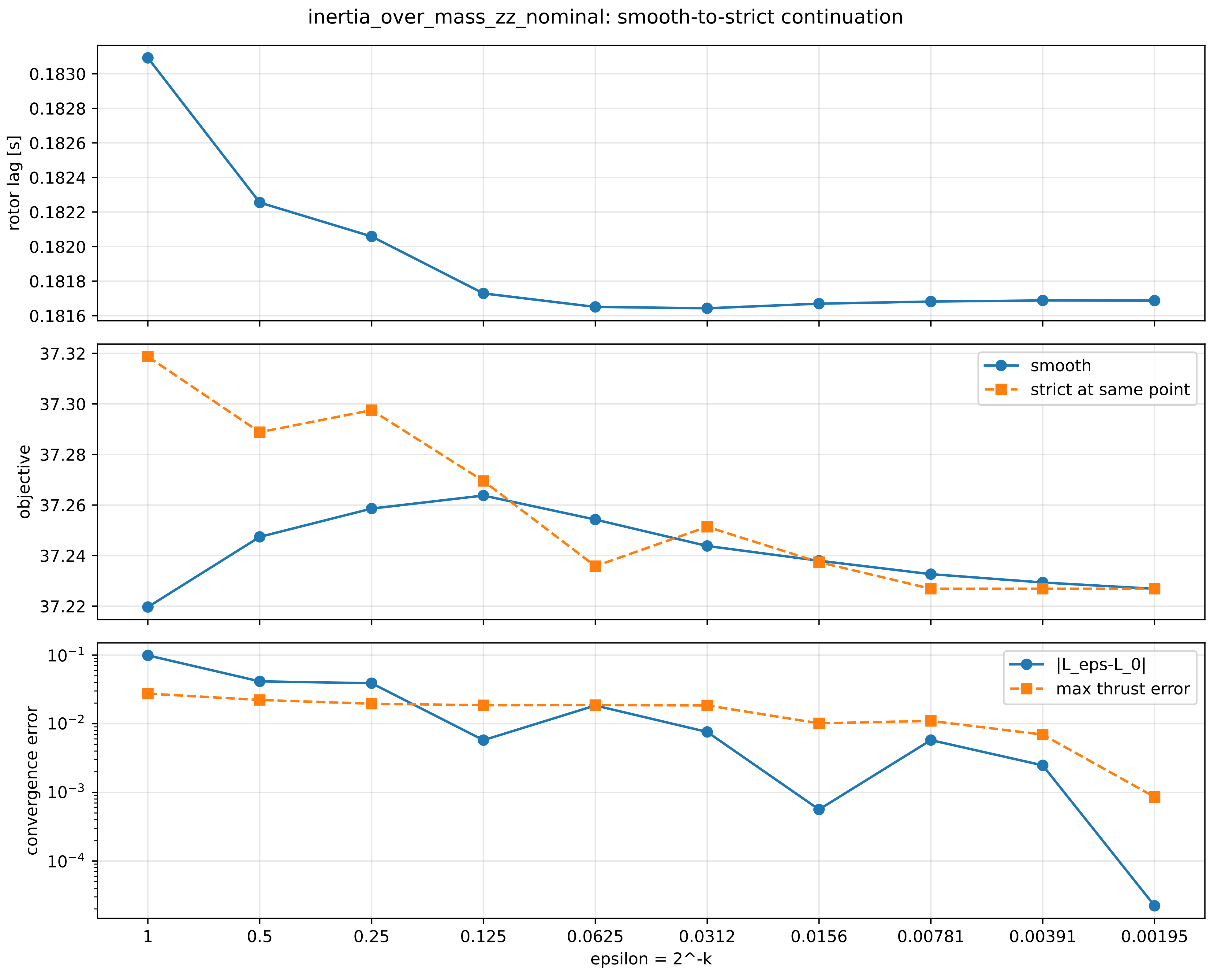


inertia_over_mass_zz_nominal: acceleration residual objective
specific acceleration [m/s^2]
angular acceleration [rad/s^2]
SG observed
model
x
y
z
time [s]



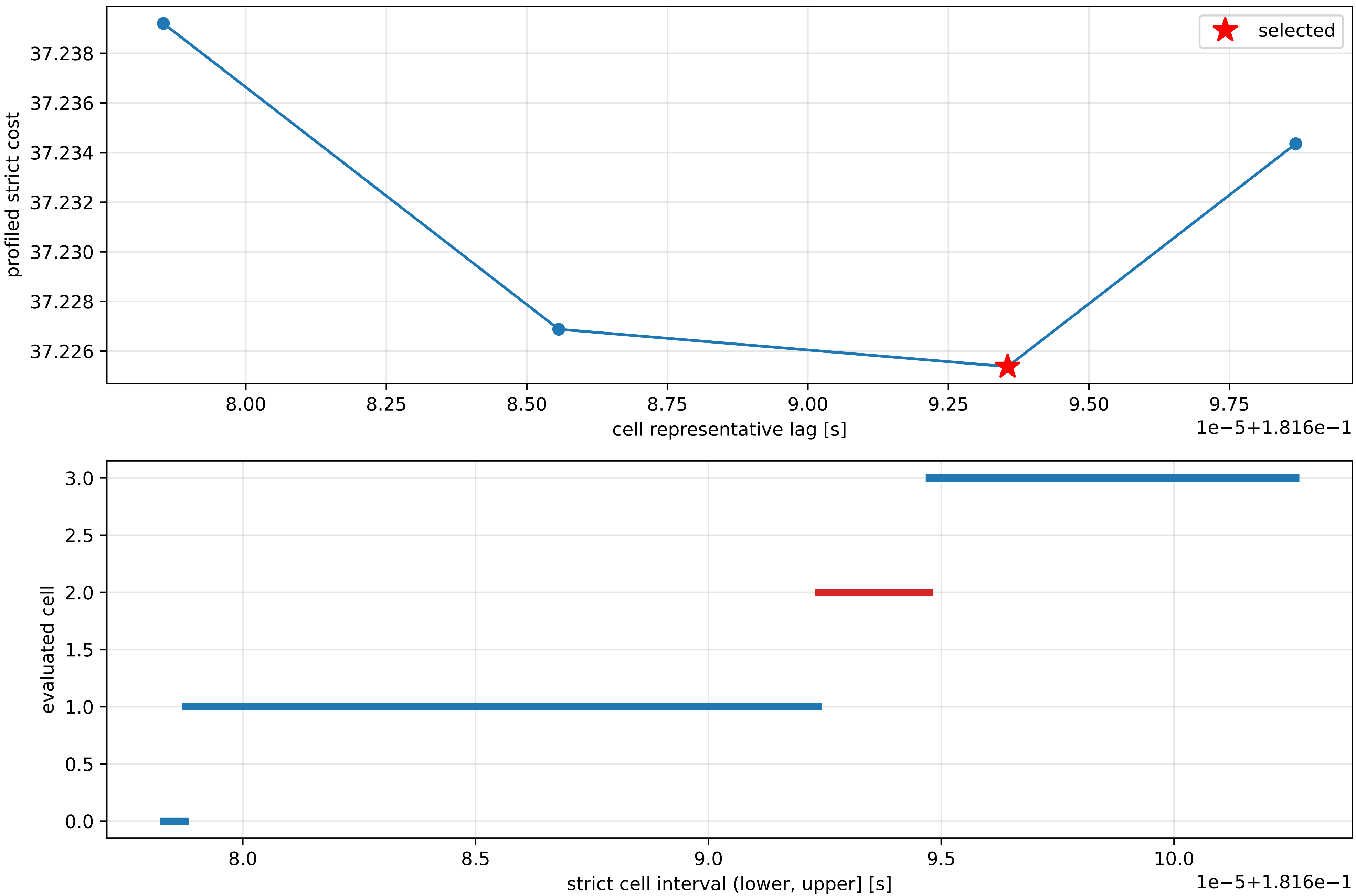
inertia_over_mass_zz_nominal: gimbal measurement audit (objective=measured_sg)
gimbal 1: RMSE=0.006345, max=0.01494 rad
gimbal 2: RMSE=0.008835, max=0.04462 rad
gimbal 3: RMSE=0.007019, max=0.01611 rad
gimbal 4: RMSE=0.006321, max=0.03642 rad
raw
SG
command replay
angle [rad]
time [s]


inertia_over_mass_zz_nominal: IMU comparison (diagnostic only)
gyro [rad/s]
specific force [m/s^2]
measured
post-fit
x
y
z
time [s]
0.10
0.05
0.00
−0.05
−0.10
0.15
−0.15
1.6
1.4
1.2
1.0
0.8
0.6
0.4
0.2
1.00
0.75
0.50
0.25
−0.25
−0.50
−0.75
9.85
9.80
9.75
9.70
9.65
9.60
9.55
0
2
4
6
8



inertia_over_mass_zz_nominal: smooth-to-strict continuation
rotor lag [s]
0.1830
0.1828
0.1826
0.1824
0.1822
0.1820
0.1818
0.1816
objective
37.32
37.30
37.28
37.26
37.24
37.22
smooth
strict at same point
convergence error
10^{-1}
10^{-2}
10^{-3}
10^{-4}
|L_eps-L_0|
max thrust error
1
0.5
0.25
0.125
0.0625
0.0312
0.0156
0.00781
0.00391
0.00195
epsilon = 2^-k

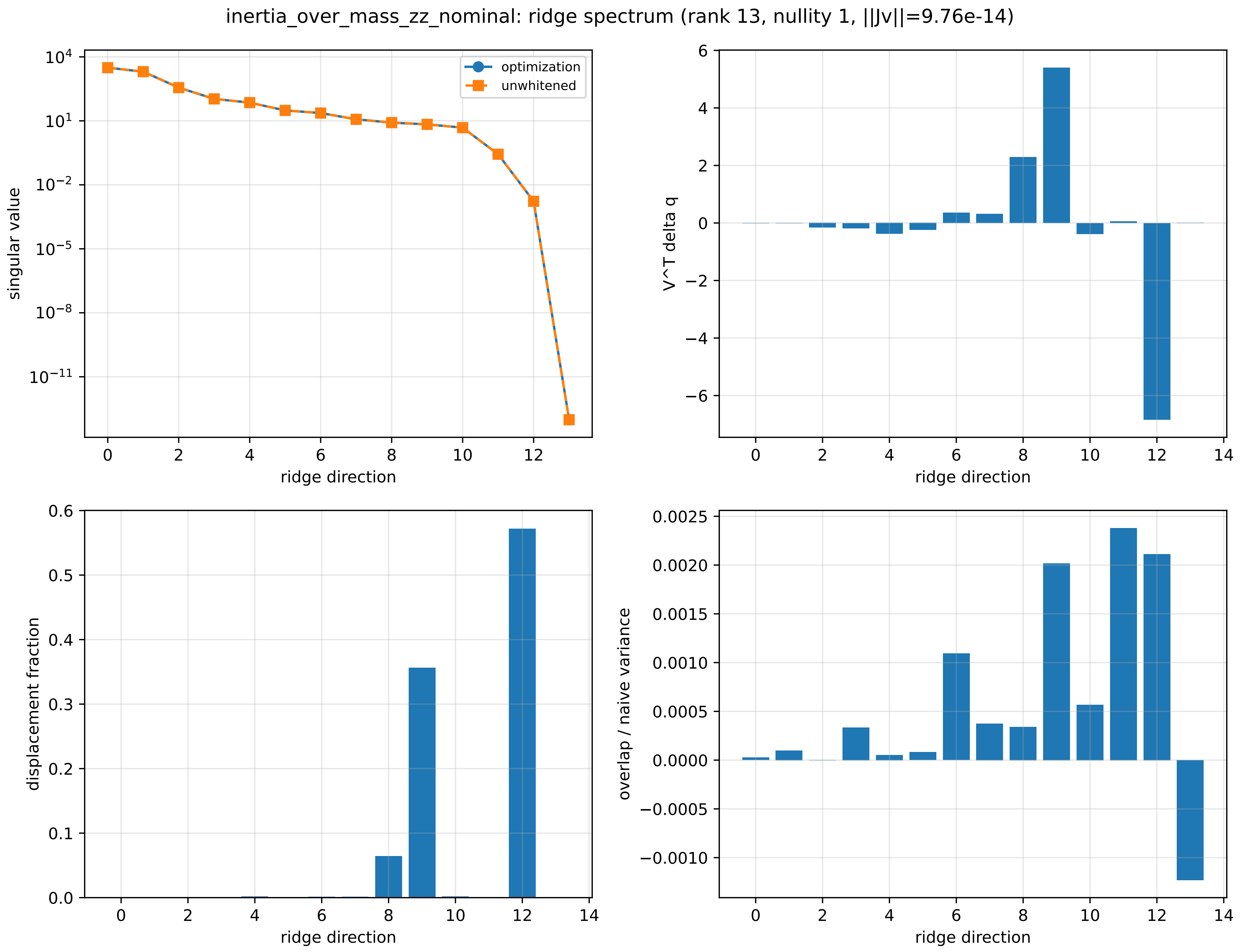
inertia_over_mass_zz_nominal: exact strict-ZOH lag cells



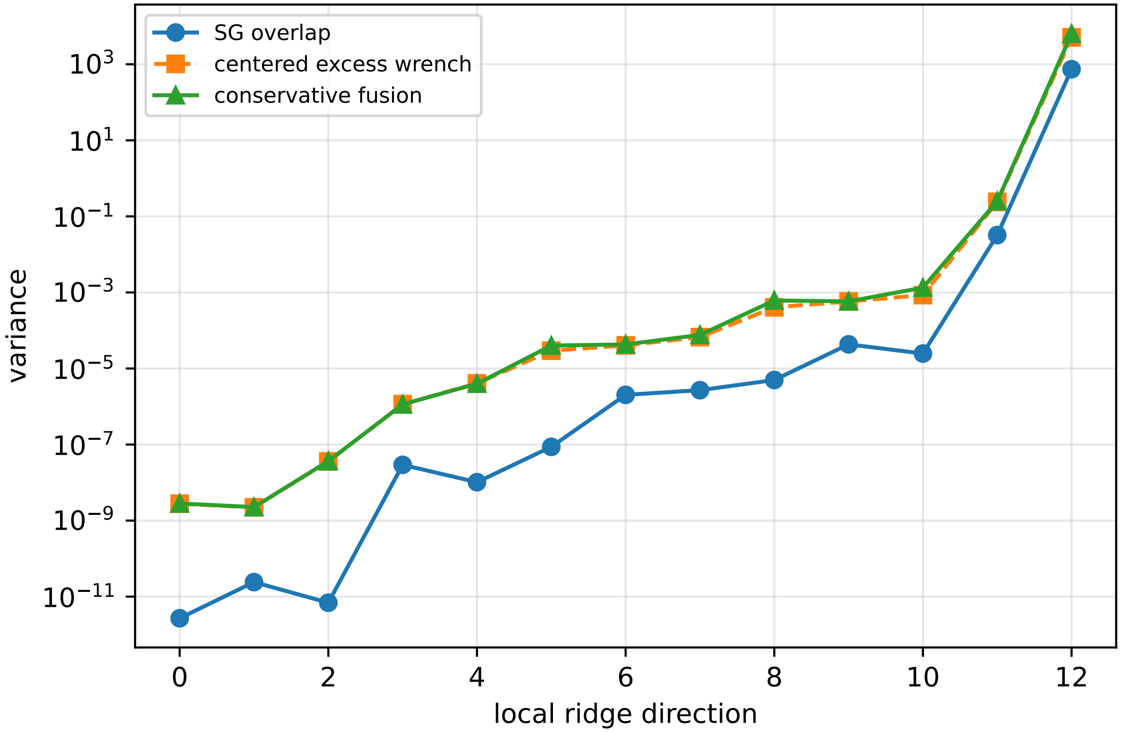
selected (0.181692362, 0.181694746] s; final smooth 0.181687483 s; data boundary=False

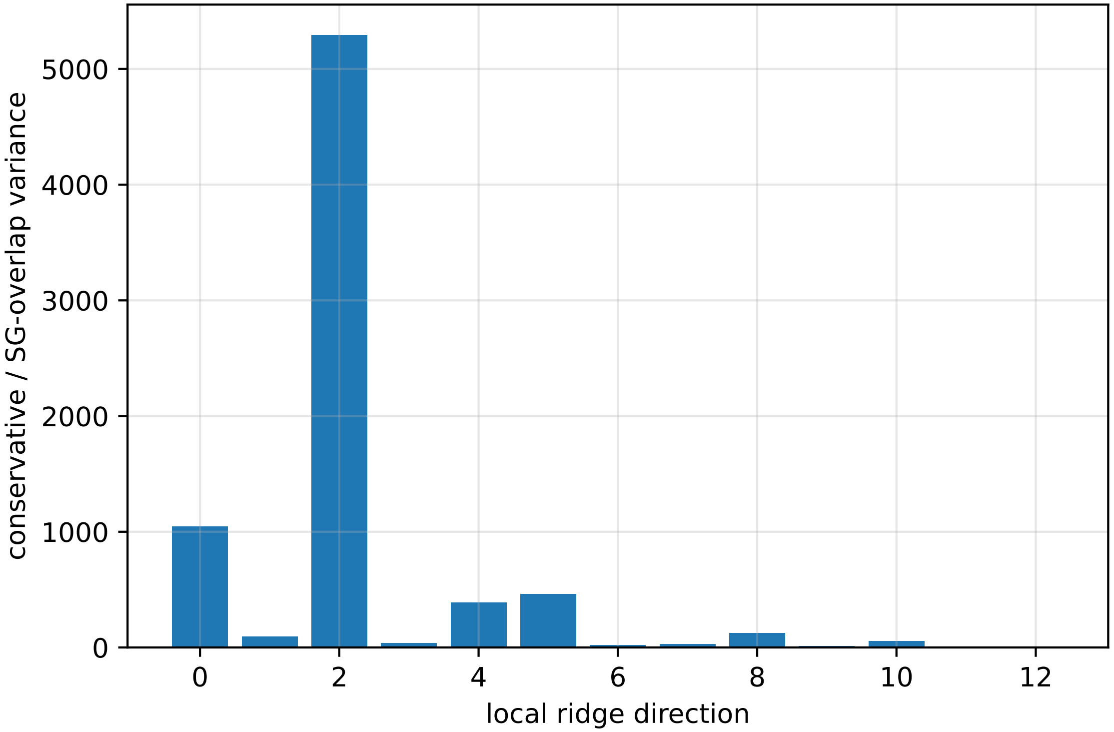
inertia_over_mass_zz_nominal: parameter estimate

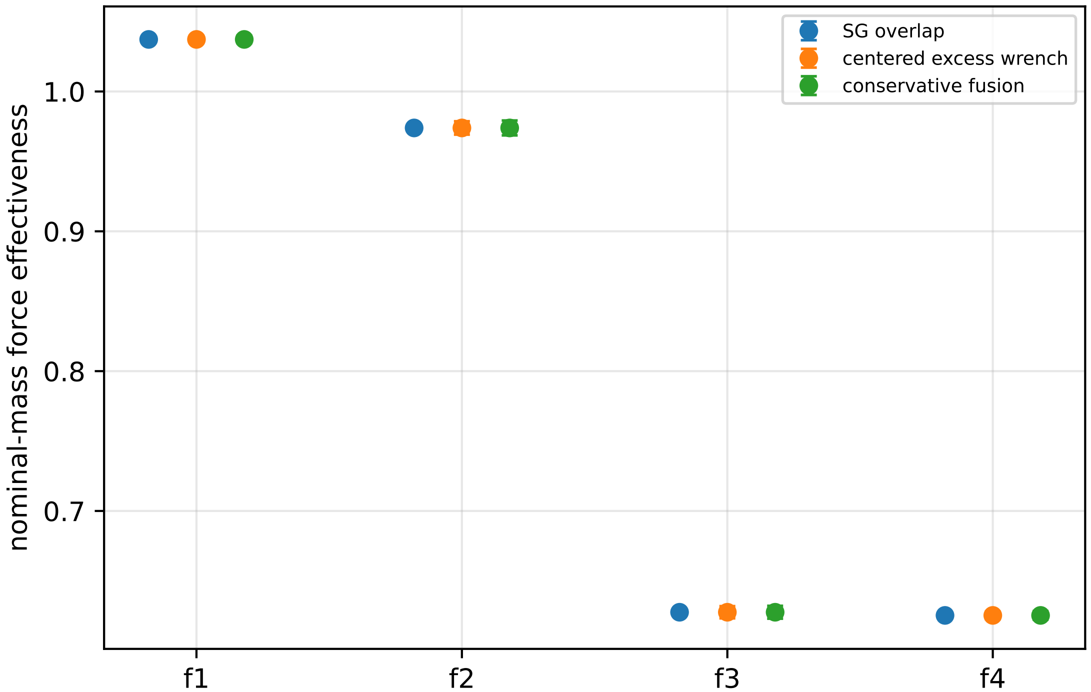
```
Case: inertia_over_mass_zz_nominal
status: completed (all stages completed)

mass [kg]: nominal 2.35159759; estimated 2.48050382
inertia [kg m^2]:
[[ 0.4037093  -0.01285773  0.00215901]
 [-0.01285773  0.27031816  0.02581007]
 [ 0.00215901  0.02581007  0.136063  ]]
CoG body [m]: [-0.01385346 -0.04778286  0.0025171 ]
force effectiveness: [1.09408496 1.02731068 0.66196854 0.6595811 ]

scale-free J/m [m^2]:
[[ 0.16275294 -0.00518352  0.00087039]
 [-0.00518352  0.10897712  0.01040517]
 [ 0.00087039  0.01040517  0.05485297]]
scale-free f/m: [0.44107369 0.41415404 0.26686858 0.2659061 ]

rotor lag cell: (0.181692362, 0.181694746] s
representative: 0.181693554 s
```


inertia_over_mass_zz_nominal: ridge spectrum (rank 13, nullity 1, ||Jv||=9.76e-14)
optimization
unwhitened
singular value
ridge direction
V^T delta q
ridge direction
displacement fraction
ridge direction
overlap / naive variance
ridge direction

inertia_over_mass_zz_nominal: Conservative fusion uncertainty

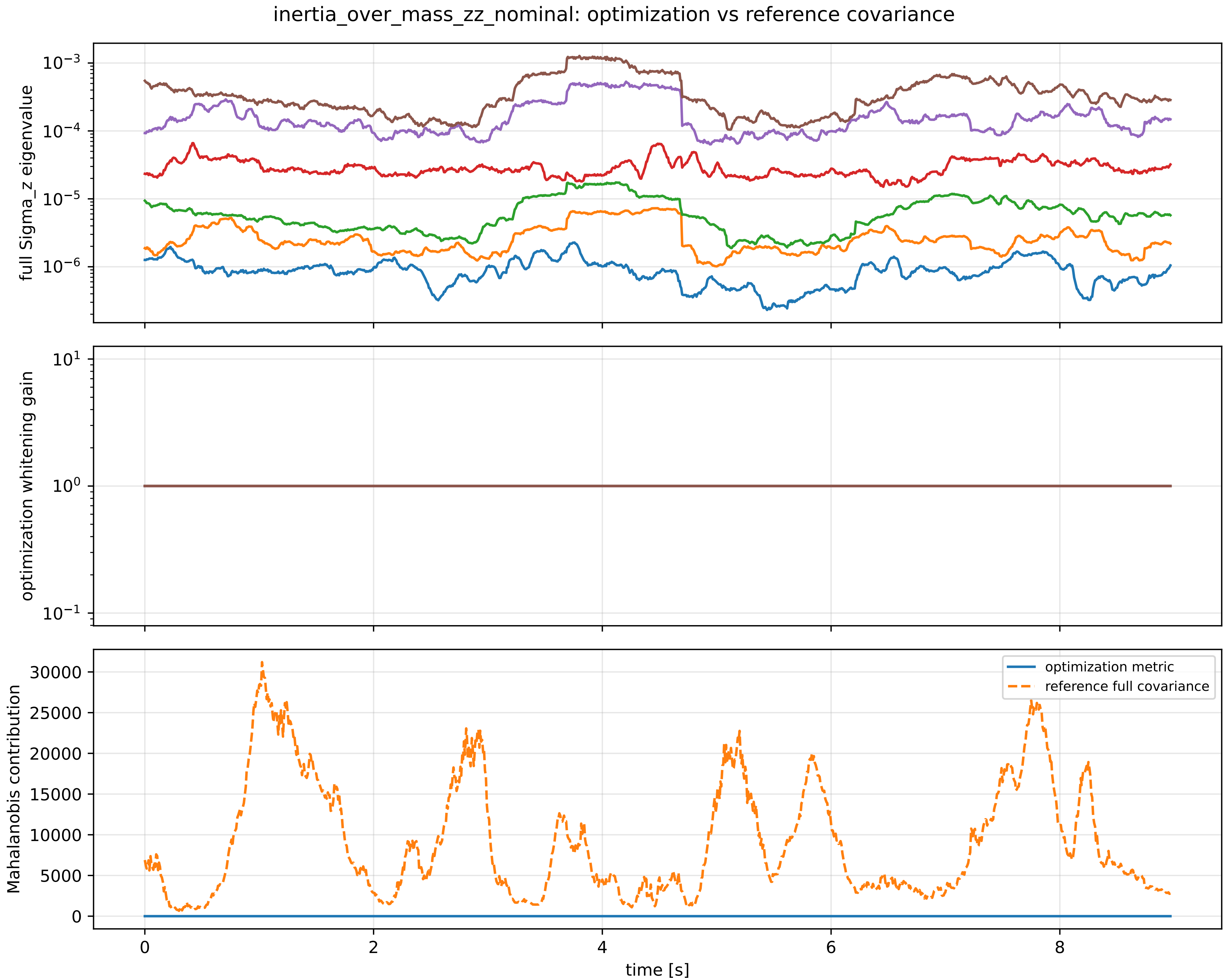


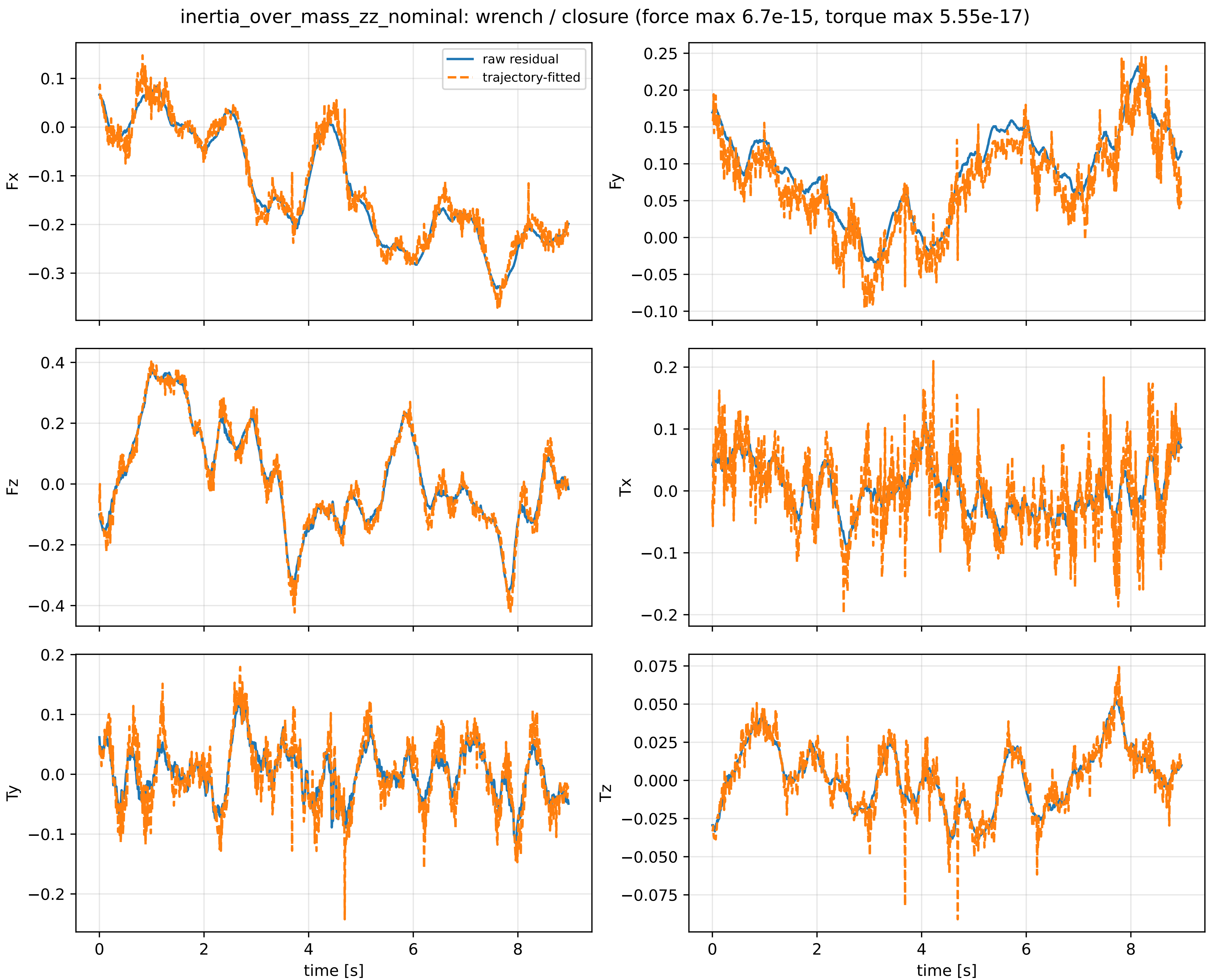





```
Top three non-gauge ambiguous directions
tr(M_res) / tr(M_SG) = 455.2
wrench-to-acceleration recovery max error = 2.7e-15
exact scale gauge ridge index = 13 (alignment 1)

index 12: variance=6331, ratio=8.542
  log_mass=-0.133; second_moment_diag_1=+0.926
  second_moment_diag_2=-0.125; second_moment_diag_3=-0.134
  second_moment_offdiag_12=-0.142; second_moment_offdiag_13=-0.000512
  second_moment_offdiag_23=+2.47e-05; cog_x=+1.62e-06
  cog_y=-6.98e-08; cog_z=+1.09e-05
  log_force_eff_1=-0.133; log_force_eff_2=-0.133
  log_force_eff_3=-0.133; log_force_eff_4=-0.133

index 11: variance=0.251, ratio=7.794
  log_mass=-0.00187; second_moment_diag_1=-0.133
  second_moment_diag_2=+0.134; second_moment_diag_3=+0.00941
  second_moment_offdiag_12=-0.982; second_moment_offdiag_13=-0.00905
  second_moment_offdiag_23=+0.00131; cog_x=+4.84e-05
  cog_y=-0.000256; cog_z=-0.000122
  log_force_eff_1=-0.00234; log_force_eff_2=+0.000803
  log_force_eff_3=-0.00519; log_force_eff_4=-0.00172

index 10: variance=0.001336, ratio=54.09
  log_mass=+0.000175; second_moment_diag_1=+0.0015
  second_moment_diag_2=+0.086; second_moment_diag_3=-0.0716
  second_moment_offdiag_12=+0.0202; second_moment_offdiag_13=-0.984
  second_moment_offdiag_23=+0.0407; cog_x=-0.0122
  cog_y=-0.000559; cog_z=-0.00158
  log_force_eff_1=+0.0358; log_force_eff_2=-0.036
  log_force_eff_3=-0.0908; log_force_eff_4=+0.075
```

The conservative fusion covariance deliberately retains the existing SG-overlap uncertainty and adds the uncentered empirical residual score second moment without subtracting the SG contribution. It is intended as a conservative fusion distribution, not as a calibrated generative noise covariance.


inertia_over_mass_zz_nominal: optimization vs reference covariance
full Sigma_z eigenvalue
10^{-3}
10^{-4}
10^{-5}
10^{-6}
optimization whitening gain
10^{1}
10^{0}
10^{-1}
Mahalanobis contribution
30000
25000
20000
15000
10000
5000
0
optimization metric
reference full covariance
0
2
4
6
8
time [s]


inertia_over_mass_zz_nominal: wrench / closure (force max 6.7e-15, torque max 5.55e-17)
raw residual
trajectory-fitted
Fx
Fy
Fz
Tx
Ty
Tz
time [s]
time [s]

inertia_over_mass_zz_nominal: optional parameter-prior audit

```
Optional physical parameter prior

The parameter prior is optional external information. The default estimator is prior-free.
name: pseudo_condition_inertia_over_mass_zz_nominal
role: pseudo_conditioning_ablation
source: /home/leus/catkin_ws/src/ProbTF-demo/ros/examples/grape-param-estim/minimal/config/priors/pseudo_conditioning/scalar/inertia_over_ma
SHA256: 0aabc66d65b6ca544bf367511021d32b9f332e43a03fe96c3ce28f09a6e7e577

data objective:  37.2253764674
prior objective: 1.62382554737e-07
total objective: 37.2253766298

This configuration uses an intentionally tight artificial standard deviation
for an ablation-style pseudo-conditioning experiment; it is not a calibrated
physical uncertainty.

factor: inertia_over_mass_zz_nominal (inertia_over_mass_m2)
  components: zz
  target: [0.05485297]
  std:    [1.e-06]
  final:  [0.05485297]
  error:  [5.69881663e-10]
  standardized residual: [0.00056988]
  factor objective: 1.62382554737e-07
```